



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	PETROCHEMICALS AND COSMETICS				
Type of Course	ELECTIVE IN MAJOR				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. A fundamental understanding of general chemistry is essential. 2. Proficiency in organic chemistry is crucial, as petrochemicals and cosmetics are primarily composed of organic compounds. 3. Knowledge of analytical chemistry techniques used for the characterization and analysis of petrochemicals and cosmetic products, such as chromatography, spectroscopy, and mass spectrometry, may be beneficial.				
Course Summary	This course aim to provide students with a comprehensive understanding of petroleum products and their purification processes, as well as the skills necessary for careers in petroleum refining, petrochemicals, quality control, and environmental compliance. Study of the various ingredients used in cosmetics, including their sources, functions, properties, and effects on the skin and hair.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	A comprehensive understanding of the petrochemical industry, including its role in the global economy, key players, and major processes involved.	U	F	Instructor-created exams / Quiz /Assignment
CO2	Understanding of the composition of petroleum, including the various hydrocarbon compounds present and their physical and chemical properties. Understanding of synthetic methods for producing	Ap	C	Class test /Assignment /Quiz

	organic compounds from hydrocarbons.			
CO3	Realizing the physical and chemical properties of different hydrocarbon fractions in crude oil, including their boiling points, densities, viscosities, and reactivities. Familiarity with the principles of distillation. Understanding of the equipment and processes used in crude oil distillation.	U	P	Assignment/ Class test
CO4	Familiarise with the diverse range of products derived from petroleum refining and the purification techniques employed to remove impurities from petroleum products.	An	C	Assignments /Seminar presentation
CO5	Explore applications of petro chemistry in various fields, such as cosmetic and perfume industry and also the various ingredients used in perfumes and cosmetics, including natural and synthetic fragrances, essential oils, emollients, surfactants, preservatives, colorants, and other functional additives.	Ap	M	Assignments /Seminar presentation
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Introduction to Petrochemistry		9	15
	1	Introduction. Petroleum. Refining of crude oil.	2	
	2	Fuels for internal combustion engines. Knocking, Octane number. Unleaded petrol. Diesel Engine and Cetane number.	3	
	3	Cracking. Thermal, Catalytic. Mechanism of cracking process.	2	
	4	Reforming Activation Gasoline. Petrochemicals.	2	
II	Hydrocarbons from Petroleum and Industrial organic synthesis		12	20
	6	Introduction. Raw materials. Saturated hydrocarbons from natural gas. Uses of saturated hydrocarbons. Unsaturated hydrocarbons – Acetylene, Ethylene, Propylene.	2	
	7	Aromatic hydrocarbons - Benzene. Toluene. Chemical processing of paraffin hydrocarbons. Chemical processing of ethylene hydrocarbons. Chemical processing of acetylene. Chemical processing of aromatic hydrocarbons.	3	

	8	Introduction to industrial organic synthesis from Petroleum. The raw materials and basic processes. Chemical process used in industrial organic synthesis.	2	
	9	Petrochemicals- Methanol. Important points. Ethanol. Important points. Rectified spirit from beer. Methylated spirit. Proof spirit. Preparation of the absolute alcohol from rectified spirit.	3	
	10	Acetaldehyde. Acetic acid. Isopropanol. Ethylene glycol. Glycerine. Acetone. Phenol. Formaldehyde. Important points.	2	
III	Composition of Petroleum Crude and Distillation of Crude Petroleum		15	25
	11	Composition of petroleum crude. Composition of the petroleum products. Isomeric compounds. Classification of petroleum crude	2	
	12	Physical Properties and Test Methods. 1. Viscosity: Other methods for finding out viscosity. Viscosity of an oil blend. Use of the figure for finding out viscosity. Viscosities of hydrocarbons. 2. Density, 3. Surface and interfacial tensions. 4. Refractive Index. 5. Flash and fire points. 6. Cloud and pour points. 7. Aniline point. 8. Diesel index. 9. Cetane number. 10. Octane number and knock characteristics.	5	
	13	Preparation of petroleum for processing. Destruction of petroleum emulsion. Electric desalting plants. Methods of petroleum distillation. Distillation of crude petroleum. .	3	
	14	Treatment of the residual liquid processing of liquid fuels such as petroleum and petroleum products. Storage tanks. Rectification columns. Cap tray or bubble tray columns.	3	
	15	Heat exchange apparatus. Steam space heaters or boilers. Condensers. Pipe furnaces. Pipelines. Fitting Compressors and pumps.	2	
IV	Petroleum products and their purification		12	30
	16	Introduction. Classification of petroleum products. Liquefied hydrocarbons, gases and fuels. Fuel oils or boiler oils. Fuel for Jet engines and gas turbine engines.	2	
	16	Lubricants, Paraffins, ceresins, petroleum. Miscellaneous petroleum products.	2	
	17	Products of petrochemical and basic organic synthesis. Dye intermediates. Lacquers. Solvents. Thinners.	2	
	18	Absorptive and adsorptive purification. Sulphuric acid purification.	3	
	19	Hydrotreating. Purification in a DC electric field. New methods of purification. De mercaptanisation.	3	
V	Perfumes and Cosmetics (Open Ended)		12	20
	Open Ended	Perfumes: Introduction. Esters. Alcohols. Ketones. Ionones. Nitromusks. Aldehydes. Diphenyl compounds. Production of natural perfumes. Flower perfume. Fruit flavours. Artificial flavours. Colognes and after shave preparation. Deodorants and Antiperspirants. Cosmetics: Introduction. Shampoos. Ingredients. Recipe. Hair dyeing. Materials used. Colour and	12	

		Curl of Hair. Creams and Lotions. Skin Chemicals. Their ingredients. Preparation and recipe. Lipsticks. Ingredients. Preparation and recipe.		
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References:

1. B. K. Sharma, Industrial Chemistry, Goel Publication, Goa.
2. N. K. Sinha, Petroleum Refining and petrochemicals,
3. John W. Hill, Chemistry for Changing times, Surjeet Publication
4. Uttam Ray Chaudhuri, "Fundamentals of Petroleum and Petrochemical Engineering", Boca Raton London New York.
5. S ukumar Maiti, "Introduction to Petrochemicals" India Book House Pvt Ltd.
6. Gabriella Baki, Kenneth S. Alexander, "Introduction to Cosmetic Formulation and Technology", Wiley.
7. Tony Curtis, David Williams, "Introduction to Perfumery", Micelle Press; 2nd edition ., 2000.

Mapping of COs with PSOs and POs:

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
C O 1		1	2				3		1		2		1
C O 2		1		1	2		3		1		1		1
C O 3						3	3		2		2		1
C O 4						3	3		2				1

C O 5						3	3		2		2		1
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Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓